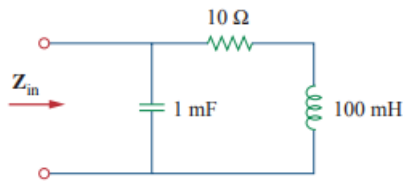


Problem

For the network in Fig. 9.66, find $\mathbf{Z}_{in}$. Let $\omega = 100$ rad/s.



Step-by-step solution

Step 1 of 4

Refer to Figure 9.66 in the text book.

Calculate the value of the reactance due to inductance.

$$X_L = j\omega L$$

Substitute 100 rad/s for ω and 100 mH for L .

$$\begin{aligned} X_L &= j(100)(100 \text{ m}) \\ &= j10 \Omega \end{aligned}$$

Therefore, the value of the reactance X_L is, $j10 \Omega$.

Step 2 of 4

Calculate the value of the reactance due to capacitance.

$$X_C = \frac{1}{j\omega C}$$

Substitute 100 rad/s for ω and 1 mF for C .

$$\begin{aligned} X_C &= \frac{1}{j(100)(1 \text{ m})} \\ &= \frac{1}{j0.1} \\ &= -j10 \Omega \end{aligned}$$

Therefore, the value of the reactance X_C is, $-j10 \Omega$.

Step 3 of 4

Draw the circuit diagram in phasor representation.

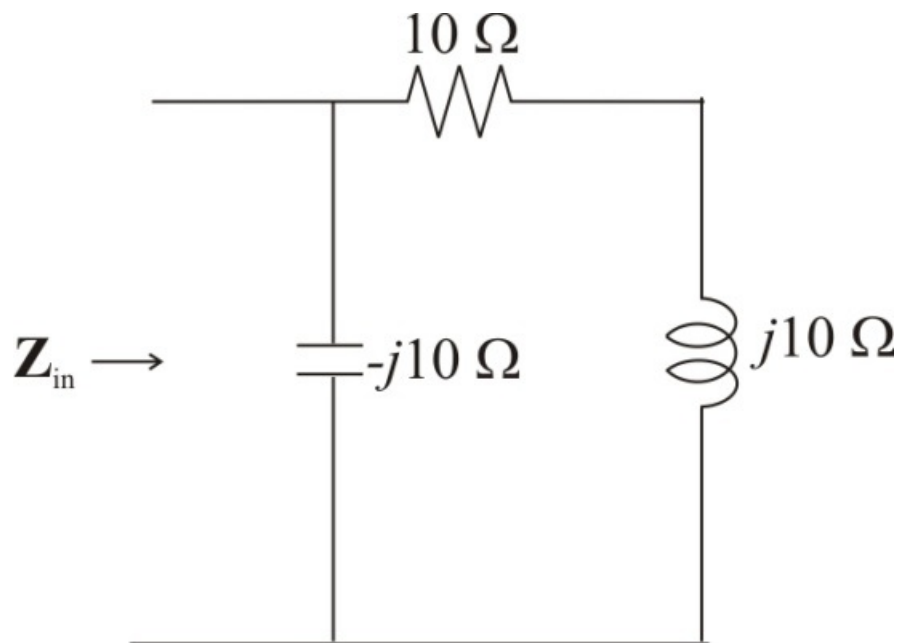


Figure 1

Step 4 of 4

Calculate the value of the impedance Z_{in} .

$$\begin{aligned} Z_{in} &= (-j10) \parallel (10 + j10) \\ &= \frac{(-j10)(10 + j10)}{-j10 + 10 + j10} \\ &= \frac{100 - j100}{10} \\ &= 10 - j10\ \Omega \end{aligned}$$

Therefore, the value of the impedance Z_{in} is, $10 - j10\ \Omega$.